

Sustainability and Business Practices
An Introduction to Project Management (MGMT-3105)

Martin Volkening

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Welcome

This book is the lecture-note companion for **MGMT-3105 – Sustainability and Business Practices**, a second-year course that introduces the philosophy of *sustainability* and the discipline of *project management* to students in electromechanical and mechanical engineering technology programs.

You are training for careers in the **automotive, food-processing, and defense** industries. In every one of those sectors you will, sooner than you expect, be handed a slice of a project: a line retrofit, a new product introduction, a validation campaign, a plant move. This course is about the structure that keeps such work on scope, on schedule, and on budget — and about doing it in a way that does not quietly borrow from the future to pay for the present.

Why a course that pairs sustainability with project management?

At first glance they look like two subjects stapled together. They are not. Every project consumes resources — materials, energy, money, human effort — and every project leaves something behind: a product, a process, a waste stream, a set of habits in the organization. **Project management is the set of tools that decides how those resources are spent, and sustainability is the lens that asks whether that spend was worth it beyond the next quarter.** A project manager who can build a work breakdown structure but cannot reason about lifecycle impact is only doing half the job.

What Is Project Management? (Association for Project Management): <https://www.youtube.com/watch?v=Jk-JwtScIlw>

What Is Sustainable Development? (FN-sambandet): <https://www.youtube.com/watch?v=7V8oFI4GYMY>

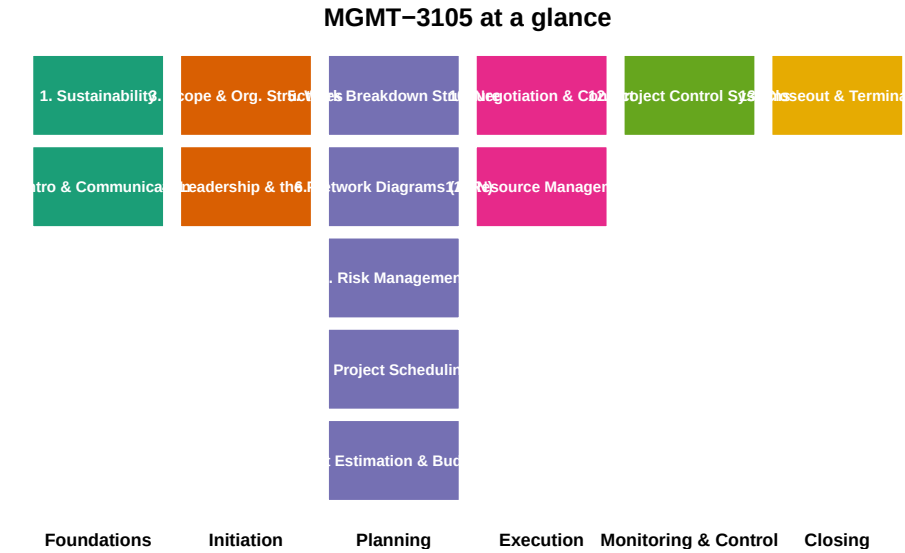


Figure 1: How the thirteen chapters of this book track the MGMT-3105 lecture sequence, grouped by project phase.

Course map

Course learning outcomes

On successful completion of MGMT-3105 you will be able to reliably:

1. Communicate the philosophy of project selection, including the Net Present Value method.
2. Communicate the theory of budgeting and scheduling.
3. Connect project-management concepts to real-world examples.
4. Identify the consequences to society of poor project planning.
5. Communicate the theory of progress management.
6. Discuss the problems of managing people and levelling allocated resources.
7. Communicate the organizational implications of closing a project.
8. Communicate negotiation theory.
9. Communicate effectively in a global environment.
10. Identify the use of sustainability in designs and in project management.

The running project

Throughout the book we return to one worked scenario, taken from the MGMT-3105 term project: your company designs and manages the launch of a **1/18-scale radio-controlled vehicle replica** for the toy market, then wins a contract to manufacture 100 000+ units for the following holiday season. You

choose the vehicle, the body and frame materials (steel, aluminium, or carbon fibre), and one of two Canadian motor/control-system suppliers. Every chapter uses a piece of that project — its scope, its WBS, its schedule, its budget, its risks, its closeout — so the tools connect to a single concrete thing rather than a dozen disconnected examples.

How the book is organized

Each chapter opens with measurable **learning objectives** and a real-world **hook**, develops the theory with plain-language explanations and LaTeX equations, and illustrates it with R-generated figures. Worked examples are shown step by step; wherever an example is a self-contained calculation, it is offered as a **live, editable R cell** you can run and modify in your browser. Collapsible **Practice** and **Discussion** blocks let you test yourself before revealing the answer, and each chapter closes with **curated YouTube videos** for a second explanation in a different voice.

Assessment (for the live course)

Table 1: MGMT-3105 grade breakdown

Component	Timing	Weight
Midterm test	Week 7	25 %
Final test	Week 14	25 %
Weekly quizzes	Weekly	12 %
Weekly case studies	Weekly	12 %
Assignment Phase A	Week 7	10 %
Assignment Phase B	Week 11	10 %
Assignment Phase C	Week 14	6 %

Quizzes and case studies require attendance. Tests are cumulative, closed-book with a handwritten aid sheet, and kit calculators are permitted. To pass, you need an overall grade of 50 % **and** completion of the mandatory work.

A note on academic integrity

Write your reports and answers in your own words. Whenever you take an idea or a passage from another author — including a classmate, a previous year’s work, or an AI tool — acknowledge it with quotation marks where appropriate and a proper citation. If you are unsure how to credit shared work, ask **before** you submit. By submitting, you put your name to the integrity of the work.

Chapter 1

Sustainability

1.1 Learning Objectives

By the end of this chapter you will be able to:

1. **Define** sustainability and sustainable design, and explain the three pillars (social, economic, ecological) and the “triple bottom line.”
2. **Identify** the four points of the sustainability platform — recycling, bio-based/biodegradable polymers, material reduction, and design for recyclability — and give an engineering example of each.
3. **Interpret** the seven plastic recycling codes and distinguish post-consumer from post-industrial recycle.
4. **Analyze** the trade-off between recyclability and biodegradability, using the corn-based PLA container case as the worked example.
5. **Calculate** the material, energy, and carbon consequences of a design decision (material substitution, wall-thickness reduction) for a production run.
6. **Evaluate** a product on whole-life cost, and explain why “cheaper to buy” and “cheaper to own” often disagree.
7. **Apply** a sustainability checklist to a project: supplier selection, contract clauses, impact assessment, circular-economy principles, and sustainability KPIs.

Before we begin: think of the last thing you threw away today

Where did its materials come from? How long did you use it before it became waste? Where is it now — landfill, recycling stream, incinerator? Could a different design decision, made years ago by an engineer you will never meet, have changed that answer? Keep the object in mind; we will come back to it.

1.2 Introduction: every gram, multiplied by a hundred thousand

Here is the situation that runs through this whole book. Your company designs and manages the launch of a **1/18-scale radio-controlled vehicle replica** for the toy market. If the prototype is accepted, you win a contract to build **100 000+ units** for the next holiday season. Early on, you must choose the body material — **steel, aluminium, or carbon fibre** — and the frame material — aluminium or carbon fibre.

That single choice is not just a strength-and-weight problem. Multiply it by 100 000 units and it becomes a decision about:

- how many tonnes of raw material leave the ground,
- how many megajoules of energy are spent turning that material into parts,
- how much scrap and packaging waste the line produces every shift,
- whether the finished toy can be recycled when a child outgrows it in three years,
- and whether your customer — a large retailer with its own public sustainability targets — will even put it on the shelf.

This is why **sustainability** belongs at the front of a project-management course, not tacked on at the end. The tools you meet in later chapters — scope, WBS, schedule, budget, risk — all decide how resources get spent. Sustainability is the question of whether that spend was worth it once you look past the launch date.

i Note

Industry snapshot. General Motors' CAMI Assembly plant in Ingersoll, Ontario — an hour from where this course is taught — runs **landfill-free**: every waste stream from daily operations is reused, recycled, or converted to energy. That is not a marketing slogan bolted onto an existing plant; it is a set of thousands of small design and process decisions of exactly the kind this chapter is about.

The Story of Stuff (Annie Leonard) - the materials economy in 20 minutes: <https://www.youtube.com/watch?v=9GorqroigqM>

1.3 What sustainability means

The most quoted definition comes from the 1987 Brundtland report and is used almost word-for-word by the U.S. Environmental Protection Agency:

Sustainability is “meeting the needs of the present without compromising the ability of future generations to meet their own needs.”

The idea is older than the word. Thomas Jefferson expressed the same principle in financial terms: **do not leave a debt for future generations to pay**. Swap “debt” for “depleted resources, degraded ecosystems, and accumulated waste” and you have the modern version.

Sustainable design applies that principle to the things engineers actually produce. Wikipedia’s working definition is useful: *the philosophy of designing physical objects, the built environment, and services to comply with the principles of social, economic, and ecological sustainability*. The intent is to *minimize negative environmental impact through skilful, sensitive design* while holding a **dynamic balance between economy and society**.

1.3.1 The three pillars and the triple bottom line

Sustainability is usually drawn as three overlapping domains. A decision is sustainable only where all three overlap: it has to be **socially** acceptable, **economically** viable, and **ecologically** sound at the same time.

```
circles <- data.frame(
  x0 = c(-0.55, 0.55, 0),
  y0 = c(0.35, 0.35, -0.6),
  r = rep(1.05, 3),
  pillar = c("Social\n(bearable)", "Economic\n(viable)", "Ecological\n(sound)"),
  fill = c("#2E86C1", "#F39C12", "#27AE60")
)

ggplot(circles) +
  ggforce::geom_circle(aes(x0 = x0, y0 = y0, r = r, fill = fill),
    alpha = 0.35, linewidth = 0.8, color = "grey30") +
  scale_fill_identity() +
  annotate("text", x = -0.95, y = 0.9, label = "SOCIAL",
    fontface = "bold", size = 4.5 * diagram_text_scale, color = "#1B4F72") +
  annotate("text", x = 0.95, y = 0.9, label = "ECONOMIC",
    fontface = "bold", size = 4.5 * diagram_text_scale, color = "#9C640C") +
  annotate("text", x = 0, y = -1.55, label = "ECOLOGICAL",
    fontface = "bold", size = 4.5 * diagram_text_scale, color = "#186A3B") +
  annotate("text", x = 0, y = 0.55, label = "bearable",
    size = 3.2 * diagram_text_scale, fontface = "italic") +
  annotate("text", x = -0.7, y = -0.35, label = "equitable",
    size = 3.2 * diagram_text_scale, fontface = "italic") +
  annotate("text", x = 0.7, y = -0.35, label = "viable",
    size = 3.2 * diagram_text_scale, fontface = "italic") +
  annotate("text", x = 0, y = -0.1, label = "SUSTAINABLE",
    fontface = "bold", size = 3.6 * diagram_text_scale, color = "grey15") +
  coord_equal() +
  labs(title = "Sustainability lives in the overlap") +
  theme_void() +
```

```
theme(
  legend.position = "none",
  plot.title = element_text(size = 14 * diagram_text_scale, face = "bold", hjust = 0
)
```

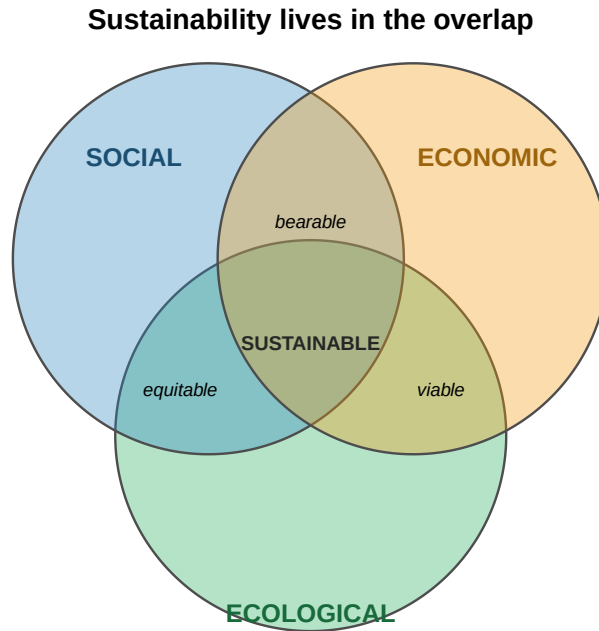


Figure 1.1: The three pillars of sustainability. Only decisions in the central overlap are genuinely sustainable; the pairwise overlaps (bearable, equitable, viable) each miss one dimension.

Discussion: which pillar does each of these decisions serve — and which does it ignore?

- A supplier offers motors 20 % cheaper, made in a plant with a poor safety record. (*economic yes; social no*)
- You switch the body to recycled aluminium, adding \$0.40 per unit. (*ecological yes; economic maybe*)
- You keep an inefficient old assembly line running because retraining staff on a new one is disruptive. (*social short-term yes; ecological and economic long-term no*)

The point of the diagram is that a “win” on one pillar is not a win. You have to check all three.

1.4 The four points of the sustainability platform

For a product engineer, sustainability comes down to four levers:

1. **Recycling** — can the material be recovered and used again?
2. **Bio-based or biodegradable polymers** — can the material come from a renewable source, or return safely to the environment?
3. **Material reduction** — can the product do its job with less material in the first place?
4. **Designing for recyclability** — is the product built so that recovery is actually practical, not just theoretically possible?

The rest of the chapter takes these one at a time, then widens out to what sustainability means when you are managing a whole project rather than designing one part.

1.4.1 Sustainable design principles

The design literature expands the four levers into a set of working principles:

Principle	What it means in practice
Low-impact materials	Non-toxic, sustainably produced or recycled materials that need less energy to process.
Energy efficiency	Choose manufacturing processes — and design products — that require less energy over their life.
Emotionally durable design	Build things people want to keep. A product kept for ten years instead of three cuts consumption and waste by more than any recycling scheme.
Design for reuse and recycling	Design for a commercial “afterlife” — but without compromising performance or service life to get there. It is a balancing act.

Notice the tension already built in: “make it last” and “make it recyclable” pull in opposite directions, and “use less material” can fight “make it durable.” Good sustainable design is the art of resolving those tensions, not pretending they do not exist.

1.5 Recycling codes

Most rigid plastics carry one of seven **resin identification codes** — the number inside the chasing-arrows triangle. The code tells a sorting facility what polymer it is looking at. The three most commonly recycled are **PET (#1)**, **HDPE (#2)**, and **PP (#5)**.

```
recycling_codes <- data.frame(
  Code = 1:7,
  Abbreviation = c("PETE / PET", "HDPE", "PVC / V", "LDPE", "PP", "PS", "OTHER"),
  Polymer = c(
    "Polyethylene terephthalate", "High-density polyethylene",
    "Polyvinyl chloride", "Low-density polyethylene",
    "Polypropylene", "Polystyrene",
    "Acrylic, polycarbonate, PLA, nylon, multi-layer"
  ),
  `Typical products` = c(
    "Drink bottles, food jars",
    "Milk jugs, detergent bottles, piping",
    "Pipe, window frames, cable insulation",
    "Films, bags, squeeze bottles",
    "Bumpers, battery cases, living hinges, tubs",
    "Foam packaging, disposable cups, CD cases",
    "Eyewear, headlamp lenses, RC-car bodies, PLA parts"
  ),
  `Recycled easily?` = c("Yes", "Yes", "Rarely", "Sometimes", "Yes", "Rarely", "Rarely"),
  check.names = FALSE
)

knitr::kable(recycling_codes, align = c("c", "l", "l", "l", "c"),
  caption = "The seven resin identification codes.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) |>
  kableExtra::column_spec(1, bold = TRUE) |>
  kableExtra::row_spec(c(1, 2, 5), background = "#EAF6EC")
```

Table 1.1: The seven resin identification codes.

Code	Abbreviation	Polymer	Typical products
1	PETE / PET	Polyethylene terephthalate	Drink bottles, food jars
2	HDPE	High-density polyethylene	Milk jugs, detergent bottles
3	PVC / V	Polyvinyl chloride	Pipe, window frames, cable
4	LDPE	Low-density polyethylene	Films, bags, squeeze bottles
5	PP	Polypropylene	Bumpers, battery cases, liv

6	PS	Polystyrene	Foam packaging, disposable cups, CD c
7	OTHER	Acrylic, polycarbonate, PLA, nylon, multi-layer	Eyewear, headlamp lenses, RC-car bodi

Two grades of recyclate matter to a manufacturer:

- **Post-industrial** (a.k.a. regrind): clean scrap from your own or a supplier's process — runners, sprues, start-up shots, off-spec parts. Easy to reuse because you know exactly what it is.
- **Post-consumer**: material that has been through a consumer's hands. It must be collected, sorted, washed, ground, and often re-pelletized, and it can carry metal, labels, food residue, and other polymers. Much harder to work with, but it is the stream that actually diverts waste from landfill.

Think about it: your RC-car body is moulded from a glass-filled ABS/PC blend. Which recycling code does it carry, and what does that tell you?

It carries **#7 (OTHER)** — any multi-polymer blend or filled material that is not one of codes 1–6 lands here. In practice, #7 is rarely recycled through curbside programs: the blend cannot be melted back into a single clean polymer. If recyclability matters for this product, that is a signal to reconsider the material *before* tooling is cut, not after.

1.6 Recyclable vs. biodegradable: a real tension

Years ago the plastics industry pursued two waste strategies at once — **recycling** and **biodegradability** — and they turned out to be partly opposed. If you make a part from recyclate and want it to last (think plastic lumber, or a pallet), the last thing you want is biodegradable material mixed in, causing the part to fail early. Contaminate a recycling stream with enough compostable polymer and the whole batch can become unusable.

The industry chose recycling. But with fossil-fuel prices and supply security back in the spotlight, **biopolymers** have returned. Some of the earliest plastics were bio-based (cellulosics, natural rubber); many new ones are built from **corn or potato starch**. **Polylactic acid (PLA)** is made from corn starch and behaves a lot like PET, but it is harder to process and degrades faster at elevated temperature.

And “biodegradable” has an asterisk. Degradation is done by **microbes**, which need **oxygen and moisture**. A modern landfill is compacted, dry, and low-oxygen — almost nothing degrades well there. You can dig up decades-old newspaper and still read it.

! Important

Case study — EcoPlast Packaging (MGMT-3105, Week 1). EcoPlast launched corn-based PLA food containers. Then: customers found them brittle in the cold and warped in heat; recyclers reported PLA contaminating conventional-plastic batches; consumers wanted eco-friendly *and* the same durability as regular plastic; and environmental groups argued the containers mostly ended up in landfill, where they did not meaningfully biodegrade — calling it “green marketing.”

Discussion (Week 1 case): was it unreasonably risky for EcoPlast to trade durability for sustainability, and what would you change?

Model answer. A product that fails in real-world use creates *more* waste, erodes consumer trust, and undermines the point of sustainability — so yes, the trade-off as executed was unreasonably risky. Consumers share *some* responsibility for unrealistic expectations, but the company had a duty to manage those expectations and to test properly before launch. Rushing to market compromised quality.

What to change: (1) real R&D and performance testing across the actual temperature and handling range before launch; (2) an accurate **caution** label (“not suitable for hot liquids”) — a danger label would be excessive; (3) consumer education on correct disposal (industrial compost, not curbside or landfill); (4) stop marketing the product as fully “sustainable” until there is evidence of an environmental benefit in real disposal conditions — otherwise the claim is exposed to a misleading-marketing challenge.

The Life Cycle of a T-Shirt (TED-Ed) - why ‘where does it go?’ is the hard question: https://www.youtube.com/watch?v=BiSYoeqb_VY

1.7 Material reduction

Reducing the material that goes into a product — at the **design stage**, by thinning walls, removing bosses, right-sizing ribs, or substituting a lighter material — pays off several times over:

- the product **costs less** (less material bought per unit),
- there is **less to recycle or dispose of** at end of life,
- **more material is left** for other uses,
- and **less energy** is spent producing *and transporting* every unit.

The catch: you cannot thin your way past the product’s functional requirements. Material reduction is enabled by better material properties, better processes, and better analysis — not by wishful thinking.

1.7.1 Worked Example 1: thinning the RC-car body

Your baseline RC-car body is injection-moulded ABS with a mass of **48 g**. Simulation says you can drop the nominal wall from 2.5 mm to 2.1 mm and still pass the drop test, cutting body mass to **41 g**. The contract run is **120 000 units**. ABS costs **\$3.10/kg**, its production embodied energy is about **90 MJ/kg**, and its cradle-to-gate carbon intensity is about **3.4 kg CO₂e/kg**.

```
mass_before_g <- 48
mass_after_g  <- 41
units         <- 120000
abs_cost_per_kg <- 3.10
abs_energy_MJ_kg <- 90
abs_co2e_kg_kg <- 3.4

saved_kg <- (mass_before_g - mass_after_g) / 1000 * units

cost_saved <- saved_kg * abs_cost_per_kg
energy_saved <- saved_kg * abs_energy_MJ_kg
co2_saved <- saved_kg * abs_co2e_kg_kg

cat("Mass removed per unit :", mass_before_g - mass_after_g, "g\n")
cat("Material saved over the run :", format(round(saved_kg), big.mark = " "), "kg\n")
cat("Material cost saved :", sprintf("$%s", format(round(cost_saved), big.mark = " ")), "\n")
cat("Production energy saved :", format(round(energy_saved), big.mark = " "), "MJ",
    sprintf("(~%s kWh)", format(round(energy_saved / 3.6), big.mark = " ")), "\n")
cat("Cradle-to-gate CO2e avoided :", format(round(co2_saved), big.mark = " "), "kg",
    sprintf("(~%.1f t)", co2_saved / 1000), "\n")
```

```
Mass removed per unit : 7 g
Material saved over the run : 840 kg
Material cost saved : $2 604
Production energy saved : 75 600 MJ (~21 000 kWh)
Cradle-to-gate CO2e avoided : 2 856 kg (~2.9 t)
```

A 7 g decision, taken once on a screen, moves almost a tonne of material and several tonnes of CO₂e once it is multiplied by the production run — before counting the lighter shipping weight.

1.7.2 Try it yourself: the material-choice decision

Now compare the three candidate body materials head to head. The live cell below holds rough figures for **steel**, **aluminium**, and **carbon-fibre composite** bodies of the mass each material would need for equal stiffness. Change the numbers, add a recycled-content option, or change the run size and see what happens to cost, energy, and carbon.

```

#| label: material-choice-compare-01
# Rough per-unit body mass for equal stiffness, and cradle-to-gate factors.
material <- c("Steel", "Aluminium", "Carbon-fibre")
mass_g    <- c(140,      70,      38)      # grams per body
cost_kg    <- c(1.20,    3.00,    28.00)    # $ per kg
energy_kg  <- c(25,     155,     300)      # MJ per kg (primary)
co2_kg     <- c(2.0,    11.0,    24.0)     # kg CO2e per kg

units <- 120000

kg_total    <- mass_g / 1000 * units
cost_total  <- kg_total * cost_kg
energy_total <- kg_total * energy_kg
co2_total   <- kg_total * co2_kg

result <- data.frame(
  Material      = material,
  Body_mass_g   = mass_g,
  Material_t    = round(kg_total / 1000, 1),
  Cost_usd      = round(cost_total),
  Energy_MWh    = round(energy_total / 3600, 1),
  CO2e_t        = round(co2_total / 1000, 1)
)
print(result, row.names = FALSE)

cat("\nLightest body:", material[which.min(mass_g)], "\n")
cat("Lowest material cost:", material[which.min(cost_total)], "\n")
cat("Lowest cradle-to-gate CO2e:", material[which.min(co2_total)], "\n")

```

What the numbers usually show — and why the “greenest” answer is not obvious

Carbon fibre gives the lightest body, which cuts shipping energy and, in a real vehicle, fuel over the whole service life. But its *production* energy and carbon per kilogram are the highest of the three, and it is the hardest to recycle (code #7, thermoset matrix). Steel is cheap and highly recycled but heavy. Aluminium sits in between and is excellent if you can get **recycled** stock — secondary aluminium uses roughly 5 % of the energy of primary.

For a *toy* that is shipped once and never driven under its own power for thousands of kilometres, the use-phase fuel saving from carbon fibre never arrives — so the decision tilts back toward recycled aluminium or steel. The lesson: **the right material depends on where the impact actually lands in the life cycle.**

1.8 Designing for recyclability

Recyclability is designed in, not hoped for. The checklist a plastics engineer can actually influence:

- Make the assembly from a **single material**, or from **compatible** materials that can be recycled together.
- Make it **easy to disassemble**.
- Prefer **snap-fits and press-fits** over screws and metal fasteners.
- **Reduce the part count**.
- **Avoid labels** (or use same-polymer labels / in-mould labelling).
- Keep **thermosets** out of thermoplastic assemblies — use thermoplastic elastomers instead of rubber seals where you can.

You may not have the final word on a design, but you can always **offer the greener alternative**. And the mindset has shifted: products used to be designed for *end of life* — last for period X, then done. Now good products are designed for **next life**: what can this material become *after* its first use?

Try it: redesign the RC-car battery door for recyclability

Current design: ABS door, two stainless M2 screws, a printed polyester label, a moulded-in TPE gasket (thermoset-like behaviour), a steel spring.

One redesign:

- Replace the two screws with a **living hinge + snap latch** moulded into the door (same ABS) — removes two dissimilar-metal fasteners and a driver operation on the line.
- Replace the adhesive polyester label with **in-mould labelling** in an ABS-compatible film, or mould the text into the part.
- Replace the steel spring with a **moulded ABS cantilever spring** if the cycle life allows.
- Keep the TPE gasket but make it a **removable press-in** part in a recycling-compatible grade, or switch to a moulded-in flexible ABS lip if the seal spec allows.

Result: the door goes from five materials and two fasteners to essentially one polymer, snaps apart by hand, and carries no adhesive.

1.9 The consumer's perspective

Sustainable design has to survive contact with real buyers. Market research is consistent on two points:

1. Environmentally conscious consumers **will pay marginally more** and **accept marginally lower fit and finish** for a product that is genuinely more environmentally friendly.

2. They have **zero tolerance** for a drop in **performance/functionality** or in **service life**. They bought the product to do a job. If a “sustainable” design wears out sooner, it just gets replaced sooner — consuming more resources and defeating the purpose.

```
tol <- data.frame(
  attribute = c("Higher price", "Cosmetic fit & finish",
               "Core performance", "Service life / durability"),
  tolerance = c(0.55, 0.45, 0.03, 0.05)
)
tol$attribute <- factor(tol$attribute, levels = rev(tol$attribute))
tol$kind <- ifelse(tol$tolerance > 0.2, "Some tolerance", "Near-zero tolerance")

ggplot(tol, aes(x = tolerance, y = attribute, fill = kind)) +
  geom_col(width = 0.65) +
  geom_text(aes(label = scales::percent(tolerance, accuracy = 1)),
            hjust = -0.15, size = 3.8 * diagram_text_scale) +
  scale_x_continuous(labels = scales::percent, limits = c(0, 0.7),
                    expand = expansion(mult = c(0, 0.05))) +
  scale_fill_manual(values = c("Some tolerance" = "#27AE60",
                              "Near-zero tolerance" = "#C0392B")) +
  labs(title = "Where consumers will --- and will not --- give ground",
       x = "Approximate share of consumers willing to accept a trade-off",
       y = NULL, fill = NULL) +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )
```

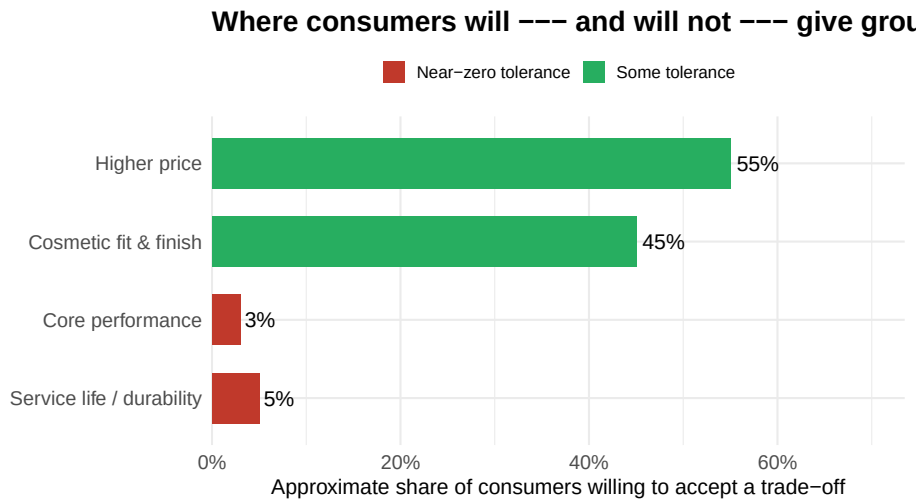


Figure 1.2: Consumer tolerance for trade-offs made in the name of sustainability: some room on price and cosmetics, essentially none on how well the product works or how long it lasts.

1.9.1 Whole-life cost: cheaper to buy vs. cheaper to own

This is where **planned obsolescence** enters. A business may have a commercial reason to limit a product's life — it protects repeat sales. Sustainable design pushes the other way: make it last. The buyer's own wallet is often on the side of durability, even when the sticker price is higher.

The live cell compares a **cheap** version of a product (low price, short life, replaced repeatedly) with a **durable** version (higher price, lasts the whole period). Change the prices, lifetimes, and horizon and watch which one wins on **total cost of ownership** — and on **units sent to landfill**.

```
#| label: whole-life-cost-01
horizon_years <- 10

# Cheap option
cheap_price    <- 18
cheap_life_yrs <- 2

# Durable option
durable_price  <- 42
durable_life_yrs <- 10

cheap_purchases <- ceiling(horizon_years / cheap_life_yrs)
durable_purchases <- ceiling(horizon_years / durable_life_yrs)
```

```

cheap_tco    <- cheap_purchases    * cheap_price
durable_tco  <- durable_purchases * durable_price

cat(sprintf("Over %d years:\n", horizon_years))
cat(sprintf("  Cheap   : %d purchases x $%d = $%d    (%d units to landfill)\n",
            cheap_purchases, cheap_price, cheap_tco, cheap_purchases))
cat(sprintf("  Durable: %d purchase(s) x $%d = $%d    (%d unit to landfill)\n",
            durable_purchases, durable_price, durable_tco, durable_purchases))

better <- if (durable_tco < cheap_tco) "Durable" else "Cheap"
cat(sprintf("\nLower total cost of ownership: %s (saves $%d)\n",
            better, abs(cheap_tco - durable_tco)))

```

Where does the cheap option actually win?

Shorten the horizon (you only need the product for a year), or make the durable version's premium extreme relative to its extra life, and “cheap” wins on cost. But the **waste count** almost always favours durable: five landfilled units versus one. Whole-life thinking means putting *both* numbers — dollars and units — in front of the decision-maker, not just the sticker price.

1.10 Sustainability in project management

Everything above was about designing a *part*. When you are running a *project*, sustainability becomes a set of management actions spread across the whole life cycle. The MGMT-3105 checklist:

Procurement and contracts

- Choose **eco-friendly and ethical suppliers**; audit them for compliance.
- Put **sustainability clauses** in contracts.
- Perform an **Environmental & Social Impact Assessment (ESIA)**.

Design and materials

- Minimize raw-material use; prioritize **recyclable, renewable, and local** resources.
- Design for **energy and water efficiency**.
- Apply **circular-economy** principles: reuse, refurbish, recycle.
- Plan the **product/service lifecycle** — what happens to it after the project ends.

Operations

- Reduce **carbon emissions** from transport and operations.
- Use **digital tools** to cut paper and travel.
- Run proper **waste management**: segregation, recycling, safe disposal.

People and governance

- Safeguard **health, safety, and well-being** of the team and the community.
- Support **social responsibility**: local hiring, fair wages, training.
- Track **sustainability KPIs**: carbon footprint, waste-reduction %, energy saved.
- Monitor suppliers and contractors; **identify and mitigate** environmental and social risks; run **sustainability audits** when needed.

1.10.1 The circular economy

The circular economy is the alternative to the linear **take — make — use — dispose** model. Instead, material is kept in use through loops of increasing size: maintain and reuse first, then refurbish and remanufacture, then recycle, with disposal as the last resort.

```
stages <- data.frame(
  label = c("Design", "Make", "Distribute", "Use", "Maintain / Reuse",
            "Refurbish / Remanufacture", "Recycle"),
  angle = seq(90, 90 - 360 + 360/7, length.out = 7) * pi / 180
)
stages$x <- cos(stages$angle)
stages$y <- sin(stages$angle)

seg <- data.frame(
  x      = stages$x,
  y      = stages$y,
  xend   = stages$x[c(2:7, 1)],
  yend   = stages$y[c(2:7, 1)]
)

ggplot() +
  geom_curve(data = seg, aes(x = x, y = y, xend = xend, yend = yend),
            curvature = -0.25, linewidth = 0.9, color = "#27AE60",
            arrow = arrow(length = unit(0.18, "cm"), type = "closed")) +
  geom_point(data = stages, aes(x = x, y = y), size = 20, color = "#EAF6EC") +
  geom_point(data = stages, aes(x = x, y = y), size = 20, shape = 1, color = "#27AE60") +
  geom_text(data = stages, aes(x = x, y = y, label = label),
            size = 3 * diagram_text_scale, fontface = "bold",
            lineheight = 0.9, color = "grey15") +
  annotate("text", x = 0, y = 0, label = "keep materials\nin use",
            size = 3.2 * diagram_text_scale, fontface = "italic", color = "grey40") +
  coord_equal(xlim = c(-1.4, 1.4), ylim = c(-1.4, 1.4)) +
  labs(title = "Design out waste; circulate materials") +
  theme_void() +
  theme(plot.title = element_text(size = 14 * diagram_text_scale,
```

```
face = "bold", hjust = 0.5))
```

Design out waste; circulate materials

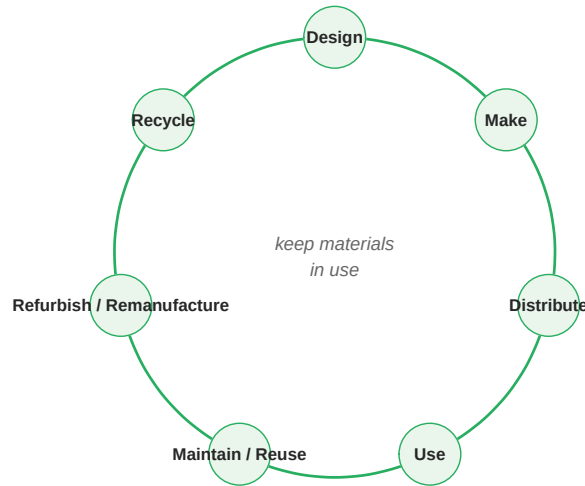


Figure 1.3: The circular economy: keep materials circulating through the tightest loop that still works. Inner loops (maintain, reuse) retain the most value; recycling is the outermost loop before disposal.

Explaining the Circular Economy (Ellen MacArthur Foundation):
<https://www.youtube.com/watch?v=zCRKvDyyHmI>

1.10.2 Try it yourself: a sustainability KPI snapshot

Sustainability only gets managed if it gets measured. The live cell computes a few of the KPIs from the checklist for the RC-vehicle project: carbon footprint per unit, waste-diversion rate, and recycled-content share. Adjust the inputs to your own project assumptions.

```
#| label: sustainability-kpi-01
units      <- 120000

# Carbon
material_co2_t  <- 71      # tonnes CO2e, cradle-to-gate materials
process_co2_t   <- 18      # tonnes CO2e, moulding + assembly energy
```

```

transport_co2_t <- 26      # tonnes CO2e, inbound + outbound freight

# Waste
process_waste_t <- 9.0     # tonnes of scrap + packaging generated
waste_diverted_t <- 8.1    # tonnes reused / recycled / to-energy

# Materials
recycled_input_t <- 15.5   # tonnes of recycled feedstock
total_input_t    <- 62.0   # tonnes of material input

total_co2_t <- material_co2_t + process_co2_t + transport_co2_t

cat(sprintf("Carbon footprint      : %.1f t CO2e total,  %.3f kg CO2e per unit\n",
            total_co2_t, total_co2_t * 1000 / units))
cat(sprintf("Waste diversion rate  : %.1f %% (target: >= 90 %%)\n",
            100 * waste_diverted_t / process_waste_t))
cat(sprintf("Landfilled waste      : %.2f t\n", process_waste_t - waste_diverted_t))
cat(sprintf("Recycled content      : %.1f %% of material input\n",
            100 * recycled_input_t / total_input_t))

status <- if (100 * waste_diverted_t / process_waste_t >= 90) "PASS" else "BELOW TARGET"
cat(sprintf("\nWaste KPI status: %s\n", status))

```

1.11 Key Takeaways

- **Sustainability** = meeting present needs without compromising future generations' ability to meet theirs. **Sustainable design** applies that across social, economic, and ecological pillars at once — a “win” on one pillar is not a win.
- The four levers for a product engineer: **recycle, bio-source/biodegrade, reduce material, design for recyclability**.
- The **seven resin codes** tell a sorter what a plastic is; PET (#1), HDPE (#2), and PP (#5) recycle well, #7 (blends/PLA/filled) rarely does. **Post-consumer** recyclate diverts real waste but is hard to process; **post-industrial** regrind is easy.
- **Recyclable and biodegradable can conflict**, and “biodegradable” does little in a dry, compacted landfill — the EcoPlast case shows what happens when that asterisk is ignored.
- A small **material-reduction** decision, multiplied by the production run, moves tonnes of material, energy, and CO2e. The “greenest” material depends on **where in the life cycle the impact lands** — use phase vs. production vs. end of life.
- Consumers give ground on **price and cosmetics** but not on **performance or durability**; **whole-life cost** usually favours the durable de-

sign on both dollars and landfill count.

- In **project management**, sustainability is a checklist across procurement, design, operations, and governance — backed by **circular-economy** loops and tracked with **KPIs** (carbon per unit, waste diversion, recycled content).

1.12 Practice Problems

Problem 1: Packaging substitution (food industry)

A food-processing client ships 2.4 million meal trays a year. The current tray is 22 g of black CPET (#1, but often rejected by sorters because black plastic is invisible to optical sorting). A proposed natural-coloured PP (#5) tray weighs 19 g, costs \$0.006/g versus \$0.005/g for the CPET, and is accepted by the client's regional recycler.

- What is the annual change in material mass and material cost?
- Give two non-cost arguments for switching, and one argument against.

Solution.

```
trays      <- 2.4e6
mass_cpet_g <- 22
mass_pp_g  <- 19
cost_cpet  <- 0.005
cost_pp    <- 0.006

mass_change_kg <- (mass_pp_g - mass_cpet_g) / 1000 * trays
cost_change    <- (mass_pp_g * cost_pp - mass_cpet_g * cost_cpet) * trays

cat("Annual material mass change :", format(round(mass_change_kg), big.mark = " "), "kg\n",
    "(a reduction)\n")
cat("Annual material cost change : ",
    sprintf("%s", format(round(cost_change), big.mark = " ")),
    " (an increase)\n", sep = "")
```

Annual material mass change : -7 200 kg (a reduction)

Annual material cost change : \$9 600 (an increase)

- About 7 200 kg less plastic per year, but roughly \$3 700/yr more spent on material because PP costs more per gram here and the 3 g saving does not fully offset it.
- For:* the natural-coloured PP is actually recovered by the recycler (black CPET usually is not), and PP trays tolerate reheating better. *Against:* higher material spend, and a tooling change is needed — capital plus a qualification run.

Problem 2: Recycled-aluminium body (automotive/toy)

Switching the RC-car body from primary to **recycled** aluminium keeps the mass at 70 g/unit but changes the cradle-to-gate energy from 155 MJ/kg to 8 MJ/kg and the carbon from 11 kg CO₂e/kg to 0.6 kg CO₂e/kg. Recycled stock costs \$0.25/kg more. Run size 120 000 units.

Estimate the energy saved, the CO₂e avoided, and the extra material cost.

Solution.

```
mass_kg    <- 70 / 1000
units      <- 120000
kg_total   <- mass_kg * units

energy_saved <- kg_total * (155 - 8)
co2_saved    <- kg_total * (11 - 0.6)
extra_cost   <- kg_total * 0.25

cat("Aluminium in the run :", format(round(kg_total), big.mark = " "), "kg\n")
cat("Production energy saved :", format(round(energy_saved), big.mark = " "), "MJ",
    sprintf("(~%s kWh)\n", format(round(energy_saved / 3.6), big.mark = " ")))
cat("CO2e avoided :", format(round(co2_saved), big.mark = " "), "kg",
    sprintf("(~%.1f t)\n", co2_saved / 1000))
cat("Extra material cost :",
    sprintf("$%s\n", format(round(extra_cost), big.mark = " ")))
```

```
Aluminium in the run : 8 400 kg
Production energy saved : 1 234 800 MJ (~343 000 kWh)
CO2e avoided : 87 360 kg (~87.4 t)
Extra material cost : $2 100
```

Roughly 1.2 million MJ (about 340 000 kWh) and 87 t of CO₂e avoided for about \$2 100 of extra material cost — one of the highest-leverage sustainability moves available on this project.

Problem 3: Design-for-recyclability audit (defense)

A defense client requires that field-returned electronic enclosures be **materially recoverable**. You are handed this enclosure: glass-filled nylon (#7) housing, 6 zinc-plated steel screws, a bonded aluminium RFI shield, a polyurethane potting compound over the PCB, and two adhesive asset-tracking labels.

List, in priority order, the four changes that would most improve material recovery, and note the functional risk of each.

Model answer.

1. **Replace the potting compound** with a removable conformal coating or a gasketed enclosure. *Risk*: reduced shock/moisture protection — must re-qualify to the environmental spec.

2. **Eliminate the bonded aluminium shield**; use a conductive coating on the nylon or a snap-in shield of the same-family metal. *Risk*: RFI performance must be re-verified.
3. **Swap the 6 steel screws** for moulded snap features or same-metal fasteners. *Risk*: fewer re-assembly cycles; captive-hardware requirement may forbid it.
4. **Drop adhesive labels** for moulded-in or laser-etched marking. *Risk*: contrast/legibility and traceability sign-off.

The housing polymer itself (#7 glass-filled) is hard to change if the structural spec depends on it — note it as a known limitation rather than a quick fix.

Problem 4 (discussion): planned obsolescence vs. sustainable design

“Sustainable design requires that our products last as long as possible. But business decisions may deliberately shorten product life to protect repeat sales — planned obsolescence. Which do you prefer, and how would you reconcile the two in a company that has to stay profitable?”

There is no single right answer — this is a position you argue. Strong answers usually land on **service and circular business models**: the manufacturer keeps revenue through repair, upgrades, refurbishment, take-back, or product-as-a-service, so “make it last” and “keep earning” stop being opposites. Weak answers just pick a side and ignore the constraint on the other.

1.13 Further Resources

- **U.S. EPA — Learn About Sustainability**: <https://www.epa.gov/sustainability/learn-about-sustainability>
- **Brundtland Report (*Our Common Future*, 1987)**: <https://sustainabledevelopment.un.org/content/documents/5987our-common-future.pdf>
- **Ellen MacArthur Foundation — Circular Economy Introduction**: <https://www.ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview>

The Story of Stuff: <https://www.youtube.com/watch?v=9GorqroigqM>

Explaining the Circular Economy (Ellen MacArthur Foundation): <https://www.youtube.com/watch?v=zCRKvDyyHmI>

What Is Sustainable Development?: <https://www.youtube.com/watch?v=7V8oFI4GYMY>

Chapter 2

Introduction to Project Management and Communication

Note

This chapter is in preparation. It will be built from **Lecture 2 – Introduction and Communication** of MGMT-3105 and the Week 2 case study (“Orion Tech’s communication dilemma”), following the same structure as Chapter 1 and Chapter 9.

2.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Define a *project*, a *program*, and *operations*, and explain what makes project work different from routine work.
- Describe the project life cycle (initiation, planning, execution, monitoring and control, closing).
- Identify the triple constraint (scope, time, cost) and its relationship to quality.
- Build a stakeholder register and a project communication plan.
- Select appropriate communication methods (formal/informal, written/verbal, push/pull) for a given audience.
- Apply active-listening techniques and recognise barriers such as selective listening and cultural/language differences.
- Explain why performance reporting (status, progress, forecasting) matters

to stakeholders.

2.2 Topic outline

1. What is a project? History and evolution of project management.
2. The project life cycle and the process groups.
3. The triple constraint and competing demands.
4. Stakeholders: identification, analysis, and engagement.
5. Communication planning: the 5W + H of a communication plan.
6. Communication methods and channels; the channel-count formula $n(n - 1)/2$.
7. Listening skills and communication barriers.
8. Performance reporting and feedback loops.
9. Running case: setting up the communication plan for the RC-vehicle project.

Chapter 3

Project Scope, Charter, and Organizational Structures

Note

This chapter is in preparation. It will be built from **Lecture 3 – Scope and Organizational Structures** of MGMT-3105 and the course's Project Charter and Project Scope templates, following the same structure as Chapter 1 and Chapter 9.

3.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Write a project charter and explain what authority it grants the project manager.
- Develop a scope statement: deliverables, acceptance criteria, exclusions, constraints, and assumptions.
- Distinguish product scope from project scope.
- Recognise and control scope creep, and run a change-control process.
- Compare functional, projectized, and matrix (weak/balanced/strong) organizational structures.
- Evaluate the trade-offs each structure imposes on authority, resource availability, and cost.

3.2 Topic outline

1. Project selection and the business case (payback, NPV preview).

2. The project charter: purpose, contents, sign-off.
3. Scope definition and the scope statement.
4. Scope creep, gold-plating, and change control.
5. Organizational structures: functional, matrix, projectized.
6. The project management office (PMO).
7. Running case: charter and scope statement for the RC-vehicle project.

Chapter 4

Leadership and the Project Manager

Note

This chapter is in preparation. It will be built from **Lecture 4 – Leadership and Project Management** of MGMT-3105, following the same structure as Chapter 1 and Chapter 9.

4.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Distinguish management from leadership in a project context.
- Describe the sources of a project manager's power (legitimate, reward, coercive, expert, referent).
- Apply situational leadership: matching style (directing, coaching, supporting, delegating) to team maturity.
- Explain the stages of team development (forming, storming, norming, performing, adjourning).
- Identify the competencies and ethical responsibilities of an effective project manager.
- Use motivation theory (Maslow, Herzberg, expectancy) to diagnose team performance issues.

4.2 Topic outline

1. The project manager's role across the life cycle.

2. Power, influence, and authority without ownership of resources.
3. Leadership styles and situational leadership.
4. Building and developing the project team.
5. Motivation and engagement.
6. Emotional intelligence and ethics.
7. Running case: staffing and leading the RC-vehicle project team.

Chapter 5

Work Breakdown Structures (WBS)

i Note

This chapter is in preparation. It will be built from **Lecture 5 – Work Breakdown Structures** of MGMT-3105, the WBS Dictionary template, and the `work-breakdown-structure.xlsx` worksheet, following the same structure as Chapter 1 and Chapter 9.

5.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Explain the purpose of a WBS and how it links scope, schedule, and budget.
- Decompose a project into deliverables, work packages, and activities.
- Apply the 100 % rule and choose an appropriate level of decomposition.
- Number a WBS and build a WBS dictionary entry for a work package.
- Distinguish a deliverable-oriented WBS from a phase-oriented or organizational (OBS) breakdown.
- Assign responsibility with a RACI / responsibility-assignment matrix.

5.2 Topic outline

1. From scope statement to WBS.
2. Decomposition: deliverables, work packages, activities.
3. The 100 % rule; how far to decompose (the 8/80 guideline).

4. WBS numbering and the WBS dictionary.
5. Control accounts and the link to cost and schedule baselines.
6. Responsibility assignment (RACI).
7. Running case: building the WBS for the RC-vehicle project.

Chapter 6

Network Diagrams and Activity-on-Node (AON)

i Note

This chapter is in preparation. It will be built from **Lecture 6 – Network Diagrams** of MGMT-3105 and the **Critical-Path-Diagram.xlsx** worksheet, following the same structure as Chapter 1 and Chapter 9.

6.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Build an activity-on-node (AON) network from a precedence table.
- Distinguish the four dependency types (FS, SS, FF, SF) and use lead and lag.
- Perform the forward pass and backward pass to find ES, EF, LS, LF.
- Calculate total float and free float and identify the critical path.
- Explain how the critical path changes when durations or dependencies change.

6.2 Topic outline

1. From WBS activities to a precedence table.
2. AON notation; predecessors and successors.
3. Dependency types, leads, and lags.
4. Forward pass: earliest start and finish.
5. Backward pass: latest start and finish.

6. Float, the critical path, and near-critical paths.
7. Running case: the AON network for the RC-vehicle project.

Chapter 7

Risk Management

i Note

This chapter is in preparation. It will be built from **Lecture 7 – Risk Management** of MGMT-3105 and the Risk Brainstorming Session Worksheet template, following the same structure as Chapter 1 and Chapter 9.

7.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Distinguish risk from uncertainty, and threats from opportunities.
- Run a structured risk-identification session and write a well-formed risk statement.
- Perform qualitative analysis with a probability-impact matrix.
- Perform quantitative analysis: expected monetary value (EMV) and decision trees.
- Select response strategies (avoid, transfer, mitigate, accept; exploit, share, enhance).
- Build and maintain a risk register and set contingency and management reserves.

7.2 Topic outline

1. What risk is, and why projects have more of it early.
2. Risk identification: brainstorming, checklists, SWOT, assumptions analysis.
3. Qualitative analysis: probability and impact scales, the P-I matrix.

4. Quantitative analysis: EMV, decision trees, sensitivity.
5. Response planning for threats and opportunities.
6. Contingency reserve vs. management reserve.
7. Monitoring: risk reviews, triggers, and the risk register.
8. Running case: the risk register for the RC-vehicle project (unproven supplier, labour negotiations, new assembly line).

Chapter 8

Project Scheduling

i Note

This chapter is in preparation. It will be built from **Lecture 8 – Project Schedule** of MGMT-3105 and the **SampleGantt better.xlsx** and **gantt-chart_L.xlsx** worksheets, following the same structure as Chapter 1 and Chapter 9.

8.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Convert a network and duration estimates into a Gantt chart and a schedule baseline.
- Apply three-point (PERT) estimating: $t_e = (o + 4m + p)/6$ and activity variance.
- Compute the probability of meeting a target date using the project standard deviation.
- Compress a schedule with crashing and fast-tracking, and find the least-cost crash sequence.
- Explain resource-limited scheduling and schedule risk.
- Read and maintain a schedule: milestones, baselines, and slippage.

8.2 Topic outline

1. From AON to a time-scaled schedule.
2. Deterministic vs. probabilistic durations; PERT.
3. Probability of on-time completion.

4. Milestones and the schedule baseline.
5. Schedule compression: crashing and fast-tracking, and their costs and risks.
6. Gantt charts and tracking Ganttts.
7. Running case: the schedule for the RC-vehicle project.

Chapter 9

Cost Estimation and Budgeting

9.1 Learning Objectives

By the end of this chapter you will be able to:

1. **List** the common sources of project cost and **classify** any cost against the four cost dichotomies (direct/indirect, fixed/variable, normal/expedited, recurring/non-recurring).
2. **Calculate** direct labour cost from a rate and hours, and **apply** an indirect-cost percentage.
3. **Select** an estimating method (ballpark, parametric, feasibility, definitive) appropriate to the project stage, and **state** its expected accuracy.
4. **Explain** how planning horizon, duration, people, structure, padding, and organizational culture affect estimate quality.
5. **Compare** top-down and bottom-up budgeting and **build** a bottom-up budget from a work breakdown structure using **activity-based costing**.
6. **Construct** a time-phased budget and **plot** the cumulative cost baseline (the S-curve).
7. **Size** a contingency reserve and **distinguish** it from a management reserve.

Before we begin: you have been asked for “a number” before you have any details

A stakeholder corners you: “*Ballpark it — what will this project cost?*” You have a one-page concept and nothing else. What do you say, and how do you say it so the number does not follow you around for the next six months as a commitment? Hold that thought.

9.2 Introduction: the number that decides whether the project happens

Back to the running project. Your company must present a budget to **Senior Management before the Scope and Charter are approved**. The project brief hands you rough ranges to work from:

Table 9.1: Selected budget ranges from the RC-vehicle project brief

Item	Range
Project Manager (salary)	\$60 000 – \$80 000
Team members (salary, each)	\$30 000 – \$45 000
Sundry / overhead (per hour)	\$8.00 – \$15.00
Motor / controller	\$100 000 – \$175 000
Carbon fibre	\$120 000 – \$200 000
Aluminium	\$50 000 – \$115 000
Tooling	\$100 000 – \$180 000
Testing	\$25 000 – \$100 000
Government approval (CSA, CE, MTO)	\$2 000 – \$10 000
Packaging / shipping	\$5 000 – \$25 000

Cost estimation is the first step in building a budget, and the budget determines whether the project is feasible at all. Two principles run through everything below:

1. The more clearly you define each cost element up front, the fewer budgeting errors you make.
2. The more accurate your initial estimate, the more your budget reflects reality — and the less likely a painful overrun later.

i Note

Industry snapshot. A defense contractor bidding a **firm-fixed-price** contract must submit a *definitive* estimate ($\pm 5\%$): if the real cost comes in higher, the contractor eats the difference. A food company scoping a new retort line at the concept stage lives with a *ballpark* ($\pm 30\%$) and revises as quotes arrive. Same tools, very different tolerance for error — driven by where in the project you are standing.

The ABCs of Project Cost Management (Adriana Girdler): <https://www.youtube.com/watch?v=iEndSB5IgQM>

9.3 Where project cost comes from

Cost management starts in the **planning phase**, when you identify the project's human and material resources and lay out a **time-phased budget** that reflects when each will be used. The common sources of cost:

- **Labour** — the project team and anyone charging time to the project.
- **Materials** — everything consumed to build the deliverable.
- **Subcontractors** — external vendors, *and* internal support departments (test lab, drafting) that cross-charge.
- **Equipment and facilities** — purchased, leased, or allocated.
- **Travel.**
- **Shipping** — inbound and outbound, normal and expedited.

9.4 The eight cost types

Every cost gets sorted into **one or more** of eight categories, arranged as four either/or pairs. A single cost usually carries **two or three** labels at once.

```
cost_types <- data.frame(
  Dichotomy = c("Direct vs. Indirect", "", "Fixed vs. Variable", "",
                "Normal vs. Expedited", "", "Recurring vs. Non-recurring", ""),
  Type = c("Direct", "Indirect", "Fixed", "Variable",
           "Normal", "Expedited", "Recurring", "Non-recurring"),
  Meaning = c(
    "Clearly generated by the project (project labour, project materials, POs).",
    "Overhead: rent, property tax, benefits, utilities, depreciation, admin. Often applied as 20%",
    "Does not change with usage (a 6-month equipment lease at a flat fee).",
    "Proportional to usage (machine-hour charges, fuel, mileage).",
    "Incurred during normal operation and working hours; includes overtime that was already scheduled.",
    "Unplanned cost to go faster: rush freight, extra hires, premium to a supplier for priority.",
    "Happens again through the project life: utilities, logistics, ongoing labour, monthly maintenance.",
    "One-time, usually at the start or end: training, prototype tooling, initial market analysis."
  )
)
knitr::kable(cost_types, align = c("l", "l", "l"),
              caption = "The four cost dichotomies (eight types).") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                             full_width = FALSE) |>
  kableExtra::column_spec(2, bold = TRUE)
```

Table 9.2: The four cost dichotomies (eight types).

Dichotomy	Type	Meaning
Direct vs. Indirect	Direct	Clearly generated by the project (project labour, project materials, POs).

	Indirect	Overhead: rent, property tax, benefits, utilities, depreciation
Fixed vs. Variable	Fixed	Does not change with usage (a 6-month equipment lease)
	Variable	Proportional to usage (machine-hour charges, fuel, materials)
Normal vs. Expedited	Normal	Incurred during normal operation and working hours
	Expedited	Unplanned cost to go faster: rush freight, extra hire
Recurring vs. Non-recurring	Recurring	Happens again through the project life: utilities, logistics
	Non-recurring	One-time, usually at the start or end: training, prototyping

9.4.1 Direct labour cost

The core calculation is simple:

$$\text{Total Direct Labour Cost} = \text{Direct Labour Rate} \times \text{Hours Spent}$$

Indirect cost is harder to pin to a task, so many organizations apply it as a **percentage of direct cost**:

$$\text{Indirect Cost} = \text{Indirect Rate} \times \text{Direct Cost}, \quad \text{Indirect Rate} \approx 20\% - 50\%$$

9.4.2 Worked Example 1: fully-burdened cost of a work package

The “Design body & frame” work package needs **1 mechanical designer for 220 h** at \$46/h and **1 junior engineer for 120 h** at \$32/h. Direct materials for prototypes are \$4 500. The company applies indirect cost at **35 %** of direct cost.

```
designer_hours <- 220; designer_rate <- 46
junior_hours  <- 120; junior_rate  <- 32
direct_materials <- 4500
indirect_rate <- 0.35

direct_labour <- designer_hours * designer_rate + junior_hours * junior_rate
direct_cost   <- direct_labour + direct_materials
indirect_cost <- indirect_rate * direct_cost
total_cost    <- direct_cost + indirect_cost

cat("Direct labour   :", sprintf("%$s", format(direct_labour, big.mark = " ")), "\n")
cat("Direct materials:", sprintf("%$s", format(direct_materials, big.mark = " ")), "\n")
cat("Direct cost     :", sprintf("%$s", format(direct_cost, big.mark = " ")), "\n")
cat("Indirect (35%)  :", sprintf("%$s", format(indirect_cost, big.mark = " ")), "\n")
cat("-----\n")
cat("Fully-burdened work-package cost:",
    sprintf("%$s", format(round(total_cost), big.mark = " ")), "\n")
```


Direct labour : \$13 960
 Direct materials: \$4 500
 Direct cost : \$18 460
 Indirect (35%) : \$6 461

Fully-burdened work-package cost: \$24 921

Practice: classify the cost (from the MGMT-3105 budgeting case study)

For each item, give **up to three** labels from: A Direct, B Indirect, C Fixed, D Variable, E Recurring, F Non-recurring, G Normal, H Expedited.

#	Cost item	Answer
1	Technician's hourly wages for project assembly	A Direct, D Variable, G Normal
2	Office rent for company headquarters	B Indirect, C Fixed, E Recurring
3	One-time purchase of specialized testing software	A Direct, F Non-recurring
4	Electricity used by production equipment	D Variable, E Recurring, B Indirect
6	Hiring extra workers to meet a tight deadline	H Expedited, A Direct
9	Rush shipment of components to meet schedule	H Expedited, A Direct
12	Cost of the project manager's salary	C Fixed, E Recurring, A Direct
17	Routine monthly maintenance of equipment	F ... (<i>trick: this is E Recurring, B Indirect</i>)
23	Equipment lease for six months (flat fee)	B Indirect, C Fixed, E Recurring
40	One-time cost for prototype tooling	A Direct, F Non-recurring, C Fixed

The pattern: **anything charged straight to the project is Direct; overhead is Indirect; anything you pay to go faster is Expedited; anything paid once is Non-recurring.** Most items are genuinely two or three of these at once.

9.4.3 The dichotomies on the running project

- **Fixed vs. variable.** The 6-month lease on the CMM is **fixed** — the fee is the same whether you measure 50 parts or 5 000. The machine-hour charge on the rented moulding press is **variable**. Leasing an expensive machine is a real PM judgement call: will the usage justify the flat fee?
- **Normal vs. expedited.** Air-freighting circuit boards from the supplier to recover a slipped date is **expedited** and **direct**. The overtime you *planned* into the schedule from day one is still **normal**.
- **Recurring vs. non-recurring.** Prototype tooling and the initial market analysis are **non-recurring**. Utilities, logistics, and the team's salaries are **recurring** — and it matters which is which the moment you build a time-phased budget.

9.5 Estimating methods and their accuracy

The method you can use depends on how much you know, which depends on how far into the project you are.

```
methods <- data.frame(
  Method = c("Ballpark / order-of-magnitude", "Comparative / parametric",
    "Feasibility", "Definitive"),
  `When` = c(
    "Very early; scarce time/resources; deciding whether to even bid.",
    "A genuinely similar past project exists, with reliable archived actuals.",
    "Scope is clear and first supplier quotes are back (common in construction).",
    "Most design is complete; major POs can be written; detail is known."
  ),
  Accuracy = c("+/- 30 %", "+/- 15 %", "+/- 10 %", "+/- 5 %"),
  check.names = FALSE
)
knitr::kable(methods, align = c("l", "l", "c"),
  caption = "Estimating methods by project stage and typical accuracy.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) |>
  kableExtra::column_spec(3, bold = TRUE)
```

Table 9.3: Estimating methods by project stage and typical accuracy.

Method	When
Ballpark / order-of-magnitude	Very early; scarce time/resources; deciding whether to even bid.
Comparative / parametric	A genuinely similar past project exists, with reliable archived actuals.
Feasibility	Scope is clear and first supplier quotes are back (common in construction).
Definitive	Most design is complete; major POs can be written; detail is known.

This is the **cone of uncertainty**: the possible range around an estimate is widest at the concept stage and narrows as the project is defined.

```

cone <- data.frame(
  stage = factor(c("Ballpark", "Parametric", "Feasibility", "Definitive"),
    levels = c("Ballpark", "Parametric", "Feasibility", "Definitive")),
  x = 1:4,
  err = c(0.30, 0.15, 0.10, 0.05)
)
point_estimate <- 1.0 # $M

ggplot(cone, aes(x = x)) +
  geom_ribbon(aes(ymin = point_estimate * (1 - err),
    ymax = point_estimate * (1 + err)),
    fill = "#5DADE2", alpha = 0.35) +
  geom_line(aes(y = point_estimate * (1 + err)), linewidth = 1, color = "#2E86C1") +
  geom_line(aes(y = point_estimate * (1 - err)), linewidth = 1, color = "#2E86C1") +
  annotate("segment", x = 0.7, xend = 4.3, y = point_estimate, yend = point_estimate,
    linetype = "dashed", color = "grey30") +
  annotate("text", x = 4.3, y = point_estimate, label = "point estimate $1.0M",
    hjust = 1, vjust = -0.7, size = 3.3 * diagram_text_scale) +
  geom_point(aes(y = point_estimate * (1 + err)), size = 2.5, color = "#1B4F72") +
  geom_point(aes(y = point_estimate * (1 - err)), size = 2.5, color = "#1B4F72") +
  geom_text(aes(y = point_estimate * (1 + err),
    label = paste0("+", scales::percent(err, accuracy = 1))),
    vjust = -0.8, size = 3.2 * diagram_text_scale) +
  geom_text(aes(y = point_estimate * (1 - err),
    label = paste0("-", scales::percent(err, accuracy = 1))),
    vjust = 1.6, size = 3.2 * diagram_text_scale) +
  scale_x_continuous(breaks = 1:4, labels = levels(cone$stage)) +
  scale_y_continuous(labels = scales::dollar_format(suffix = "M", big.mark = " ")) +
  labs(title = "The cone of uncertainty", x = NULL, y = "Estimated project cost") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )

```

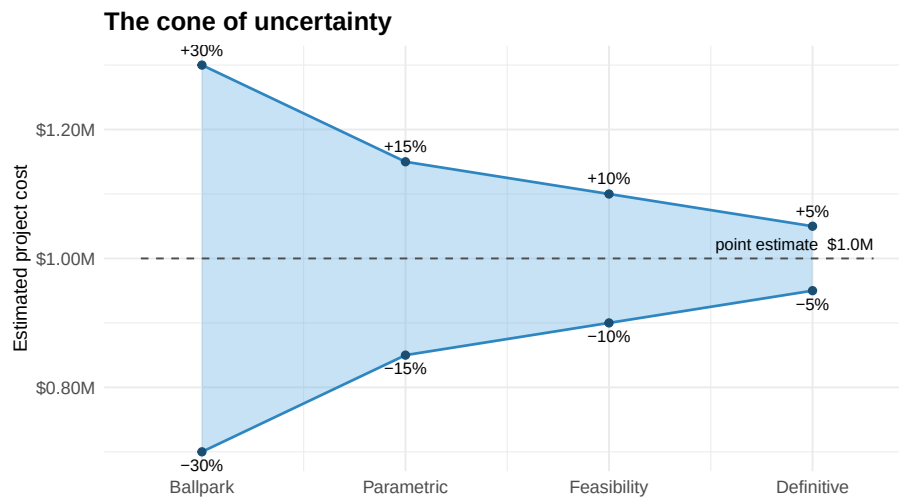


Figure 9.1: The cone of uncertainty. Early estimates can be off by +/- 30 %; the range narrows to +/- 5 % only once most design decisions are locked. The same \$1.0M point estimate means very different things at each stage.

9.5.1 Try it yourself: parametric estimate and the accuracy band

A **parametric** estimate uses a cost-estimating relationship (CER) from history — for example, “past RC-vehicle projects cost about \$14.50 per assembled part plus \$85 000 of fixed project cost.” The live cell builds the estimate and shows the band each method would put around it.

```
#| label: parametric-estimate-09
units      <- 120000
cost_per_unit_CER <- 14.50      # $/unit, from archived projects
fixed_project_cost<- 85000     # $, setup / PM / approvals

point_estimate <- units * cost_per_unit_CER + fixed_project_cost

bands <- data.frame(
  method   = c("Ballpark", "Parametric", "Feasibility", "Definitive"),
  accuracy = c(0.30,      0.15,      0.10,      0.05)
)

bands$low  <- round(point_estimate * (1 - bands$accuracy))
bands$high <- round(point_estimate * (1 + bands$accuracy))

cat(sprintf("Parametric point estimate: %s\n\n",
            format(round(point_estimate), big.mark = " ")))
for (i in seq_len(nrow(bands))) {
```

```

cat(sprintf("%-11s (+/- %2.0f%%):  %s  ...  %s\n",
           bands$method[i], bands$accuracy[i] * 100,
           format(bands$low[i], big.mark = " "),
           format(bands$high[i], big.mark = " ")))
}

```

Why quote a range at all — why not just give “the number”?

Because a single number reads as a **commitment**. If you say “\$1.8M” at the concept stage and the definitive estimate later lands at \$2.0M, you have “overrun by \$200k” — even though \$2.0M was always inside the $\pm 30\%$ ballpark band. Quoting “\$1.3M–\$2.3M, ballpark” sets the expectation honestly and buys you room to refine. A clear distinction between an **estimate** and a **commitment** is one of the most important things a PM can establish early.

9.6 What makes estimates good or bad

Factor	Effect on estimate quality
Planning horizon	Estimates of near-term work are close to 100 % accurate; accuracy falls the further out the task sits.
Project duration	Long projects and new technology add uncertainty, especially if the scope statement is weak.
People	The closer the estimator’s experience matches the work, the better. Teams that have worked together before form faster. Staff turnover hurts.
Structure	Dedicated teams deliver faster but tie up people full-time. Matrix organizations are more cost-efficient but slower.
Padding	Estimators add a safety margin. If <i>everyone</i> pads a little, the project total is badly overstated and stops reflecting reality.
Organizational culture	Some cultures reward padding; others prize accuracy. The consequences an estimator faces for a miss shape the number they give.
Other	Equipment downtime, sickness, vacation, holidays, legal limits, and project priority all leak into the estimate.

9.6.1 The cost of padding

Padding feels prudent per task. Watch what it does across a whole project.

```
set.seed(2025)
n_tasks <- 40
true_mean <- 40 # hours, honest most-likely per task
# Honest estimates vary around the truth; padded estimates add 15% on top.
honest <- rnorm(n_tasks, true_mean, 6)
padded <- honest * 1.15

df <- data.frame(
  task = 1:n_tasks,
  honest_cum = cumsum(honest),
  padded_cum = cumsum(padded)
)

ggplot(df, aes(x = task)) +
  geom_line(aes(y = honest_cum, color = "Honest estimate"), linewidth = 1.1) +
  geom_line(aes(y = padded_cum, color = "Every task padded 15%"), linewidth = 1.1) +
  annotate("text", x = n_tasks, y = df$padded_cum[n_tasks],
    label = paste0("+", round(100 * (df$padded_cum[n_tasks] /
      df$honest_cum[n_tasks] - 1)), "% on the total"),
    hjust = 1, vjust = -0.6, size = 3.4 * diagram_text_scale) +
  scale_color_manual(values = c("Honest estimate" = "#27AE60",
    "Every task padded 15%" = "#C0392B")) +
  labs(title = "Per-task padding compounds into a meaningless total",
    x = "Cumulative task count", y = "Cumulative estimated hours",
    color = NULL) +
  theme_minimal() +
  theme(
    legend.position = "top",
    plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
    axis.title = element_text(size = 11 * diagram_text_scale),
    axis.text = element_text(size = 10 * diagram_text_scale)
  )
```

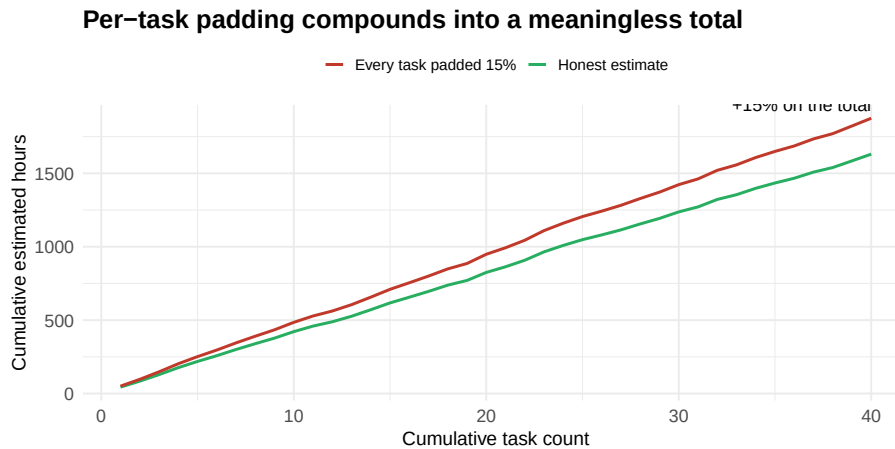


Figure 9.2: A 15 % ‘just to be safe’ pad added to every one of 40 tasks does not add 15 % to the project — it adds 15 % to the mean, but the padded total sits far above any realistic outcome because the pads do not all get used.

The fix is not “ban all safety margin” — it is to **remove padding from the work packages** and hold a single, visible **contingency reserve** at the project level (covered below).

9.6.2 Guidelines for estimating (cost and time)

- **Responsibility:** the person who will do the task makes the estimate (except for highly specialised work). It uses their experience and builds their buy-in.
- **Use several estimators:** independent estimates from knowledgeable people, reconciled, remove the extremes.
- **Assume normal conditions:** normal hours, normal resource levels — and write the assumptions down.
- **Define the time units** and the estimating base.
- **Estimate each task independently;** be careful with chains of dependent tasks, and keep one estimator per task.
- **Keep contingencies out of the work packages.** Contingency is a separate, managed fund.

9.7 Budgeting: top-down and bottom-up

The budget and the schedule are built **in tandem** — the budget decides whether the schedule’s milestones are reachable.

```
approaches <- data.frame(
  Aspect = c("How it starts", "Advantages", "Disadvantages"),
```

```

`Top-down` = c(
  "Senior management sets the overall budget and major work-package figures from experience",
  "Management aggregate estimates are often quite accurate; imposes budget discipline",
  "Creates friction: lower levels must fit management's fixed total. Turns into a zero-sum game",
),
`Bottom-up` = c(
  "Start from the WBS; apply direct and indirect cost to each activity; sum up to work package",
  "Forces detailed planning; improves PM--functional-manager communication; gives managers control",
  "Reduces management's control to oversight rather than initiation; can misalign priorities",
),
check.names = FALSE
)
knitr::kable(approaches, align = c("l", "l", "l"),
             caption = "Top-down vs. bottom-up budgeting.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                           full_width = FALSE) |>
  kableExtra::column_spec(1, bold = TRUE)

```

Aspect	Top-down
How it starts	Senior management sets the overall budget and major work-package figures from experience
Advantages	Management aggregate estimates are often quite accurate; imposes budget discipline
Disadvantages	Creates friction: lower levels must fit management's fixed total. Turns into a zero-sum game

9.8 Activity-based costing (ABC)

ABC treats a project as a series of **activities**, each of which **consumes resources**. Cost the activities and you have costed the project. Four steps:

1. Identify the activities that consume resources and assign costs to them.
2. Identify the **cost drivers** for each activity (personnel, materials, machine time...).
3. Compute a **cost rate per driver unit** (e.g. labour \$/hour).
4. **Assign cost** = rate $\times$ units the project uses. (*Example: an engineer at \$60/h working 80 h = \$4 800.*)

9.8.1 A preliminary ABC budget

Splitting each activity into its **direct** and **overhead** components:

```

abc <- data.frame(
  Activity = c("Survey", "Design", "Clear site", "Foundation", "Framing", "Plumb & w

```



```

  Direct    = c(3500, 7000, 3500, 6750, 8000, 3750),
  Overhead  = c(500, 1000, 500, 750, 2000, 1250)
)
abc$Total <- abc$Direct + abc$Overhead
abc <- rbind(abc, data.frame(Activity = "TOTAL",
                             Direct = sum(abc$Direct),
                             Overhead = sum(abc$Overhead),
                             Total = sum(abc$Total)))

knitr::kable(abc, align = c("l", "r", "r", "r"),
              format.args = list(big.mark = " "),
              caption = "Preliminary ABC budget: direct vs. overhead by activity.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                             full_width = FALSE) |>
  kableExtra::row_spec(nrow(abc), bold = TRUE)

```

Table 9.5: Preliminary ABC budget: direct vs. overhead by activity.

Activity	Direct	Overhead	Total
Survey	3 500	500	4 000
Design	7 000	1 000	8 000
Clear site	3 500	500	4 000
Foundation	6 750	750	7 500
Framing	8 000	2 000	10 000
Plumb & wire	3 750	1 250	5 000
TOTAL	32 500	6 000	38 500

9.8.2 Try it yourself: cost an activity, then a variance

The live cell costs a single activity from its drivers, then compares planned vs. actual to get the **variance**. Change the rates, hours, or actuals.

```

#| label: abc-activity-variance-09
# --- Step 3 & 4: cost the "Assemble & test prototype" activity from drivers ---
labour_rate <- 38; labour_hours <- 160
machine_rate <- 22; machine_hours <- 45
materials <- 2600
overhead_rate <- 0.30 # 30% of direct

direct <- labour_rate * labour_hours + machine_rate * machine_hours + materials
overhead <- overhead_rate * direct
planned <- direct + overhead

```

```

cat(sprintf("Direct   : $%s\n", format(round(direct), big.mark = " ")))
cat(sprintf("Overhead : $%s\n", format(round(overhead), big.mark = " ")))
cat(sprintf("Planned activity cost: $%s\n\n", format(round(planned), big.mark = " ")))

# --- Variance: what actually happened ---
actual <- 11800
variance <- planned - actual          # positive = under budget (favourable)
pct <- 100 * variance / planned

verdict <- if (variance >= 0) "UNDER budget (favourable)" else "OVER budget (unfavourable)"
cat(sprintf("Actual   : $%s\n", format(actual, big.mark = " ")))
cat(sprintf("Variance : $%s (%.1f%%) -> %s\n",
            format(round(variance), big.mark = " "), pct, verdict))

```

Warning

Common mistake. Contingency is **not** built into an ABC budget. ABC costs the activities as they are planned to run; the contingency reserve is added *afterward*, on top of the total.

9.9 The time-phased budget and the S-curve

A **time-phased budget** spreads each activity's cost across the calendar periods when it will actually be spent. Summing down the columns gives **monthly planned spend**; the running total is the **cumulative cost baseline** — the S-curve you will track the project against in Chapter 12.

```

phased <- data.frame(
  month = factor(c("Jan", "Feb", "Mar", "Apr", "May"),
                 levels = c("Jan", "Feb", "Mar", "Apr", "May")),
  Survey      = c(4000, 3000, 0, 0, 0),
  Design      = c(0, 5000, 8000, 0, 0),
  ClearSite   = c(0, 1000, 4000, 2000, 0),
  Foundation  = c(0, 0, 3000, 4000, 0),
  Framing     = c(0, 0, 0, 3000, 4000),
  PlumbWire   = c(0, 0, 0, 0, 2000)
)
phased$monthly <- rowSums(phased[, -1])
phased$cumulative <- cumsum(phased$monthly)

scale_factor <- max(phased$cumulative) / max(phased$monthly)

ggplot(phased, aes(x = month)) +
  geom_col(aes(y = monthly), fill = "#5DADE2", width = 0.6) +

```

```

geom_line(aes(y = cumulative / scale_factor, group = 1),
          color = "#C0392B", linewidth = 1.2) +
geom_point(aes(y = cumulative / scale_factor), color = "#C0392B", size = 3) +
geom_text(aes(y = cumulative / scale_factor,
              label = scales::dollar(cumulative, big.mark = " ")),
          vjust = -1, size = 3.2 * diagram_text_scale, color = "#C0392B") +
scale_y_continuous(
  name = "Planned spend this month",
  labels = scales::dollar_format(big.mark = " "),
  sec.axis = sec_axis(~ . * scale_factor, name = "Cumulative baseline (S-curve)",
                      labels = scales::dollar_format(big.mark = " "))
) +
labs(title = "Time-phased budget and cumulative cost baseline", x = NULL) +
theme_minimal() +
theme(
  plot.title = element_text(size = 14 * diagram_text_scale, face = "bold"),
  axis.title = element_text(size = 11 * diagram_text_scale),
  axis.text = element_text(size = 10 * diagram_text_scale)
)

```

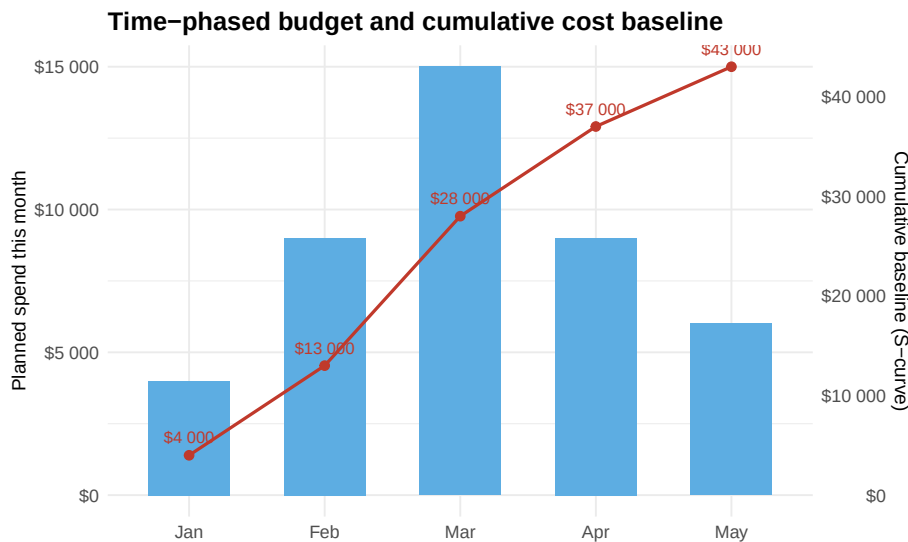


Figure 9.3: Time-phased budget for a six-activity project. Bars are planned spend per month; the line is the cumulative baseline (the S-curve). Spend is slow at first, steepest mid-project, and tapers at the end.

Matching the cost baseline to the schedule baseline is what lets you set **milestones for both schedule and budget** and later tell whether a project that has spent 60 % of its money is 60 %, 40 %, or 80 % done.

9.10 Contingency reserves

A **budget contingency** is extra money set aside to cover the uncertainty that every estimate carries — because even the best budget is only an estimate and something unforeseen always happens. Key points:

- Added **after all costs are identified**, as a cushion on top of the calculated total (commonly **10–15 %**; the MGMT-3105 Costs Chart uses **15 %**).
- **Not** included in an ABC budget — it sits on top.
- The **PM controls a project contingency fund**; separately, **top management holds a management reserve** for genuinely unknown-unknown events.

Why stakeholders push back: the percentage can look arbitrary, or feel like being asked to pay for the vendor's poor cost control. Decide early — and write into the scope statement — whether contingency applies across **all** tasks or is held for the tasks **critical** to project success.

```
reserves <- data.frame(
  Feature = c("Covers", "Owned by", "Part of the cost baseline?", "Typical size"),
  `Contingency reserve` = c(
    "Identified risks / \"known-unknowns\" (small scope adjustments, iteration, Murphy",
    "Project manager.",
    "Yes --- inside the baseline.",
    "10--15 % of estimated cost (project-dependent).",
  ),
  `Management reserve` = c(
    "Unidentified risks / \"unknown-unknowns.\",",
    "Senior management / sponsor.",
    "No --- outside the baseline; released by management.",
    "Set by organizational policy."
  ),
  check.names = FALSE
)
knitr::kable(reserves, align = c("l", "l", "l"),
  caption = "Contingency reserve vs. management reserve.") |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) |>
  kableExtra::column_spec(1, bold = TRUE)
```

Table 9.6: Contingency reserve vs. management reserve

Feature	Contingency reserve
Covers	Identified risks / "known-unknowns" (small scope adjustments, iteration, Murphy)
Owned by	Project manager.

Part of the cost baseline?	Yes — inside the baseline.
Typical size	10–15 % of estimated cost (project-dependent).

9.10.1 Try it yourself: build the RC-vehicle project budget

The live cell builds a rough **bottom-up** budget from the project brief's ranges (using a pick inside each range), rolls up labour and materials, and adds a contingency. Change the picks or the contingency rate and watch the total — then check it against what Senior Management is likely to approve.

```
#| label: rc-project-budget-09
# --- Labour (annualised salaries prorated to a 7-month project = 7/12) ---
months <- 7
pm_salary      <- 72000
n_team         <- 4
team_salary_each <- 38000
sundry_per_hour <- 11
sundry_hours   <- 1800          # total support hours booked to the project

labour <- (pm_salary + n_team * team_salary_each) * months / 12 +
  sundry_per_hour * sundry_hours

# --- Materials (one pick inside each brief range) ---
materials <- c(
  chassis_casting = 55000,
  fasteners       = 9000,
  motor_controller = 140000,
  carbon_fibre    = 0,          # not selected this time
  aluminium       = 85000,
  welding         = 25000,
  circuit_boards  = 32000,
  gov_approval    = 6000,
  testing         = 60000,
  tooling         = 150000,
  safety_equipment = 50000,
  packaging_shipping = 15000
)
material_total <- sum(materials)

subtotal      <- labour + material_total
contingency_r <- 0.15
contingency   <- contingency_r * subtotal
grand_total   <- subtotal + contingency

cat(sprintf("Labour subtotal      : $%s\n", format(round(labour),
big.mark = " ")))
```

```

cat(sprintf("Material subtotal      : %s\n", format(round(material_total),big.mark = "
cat(sprintf("Subtotal              : %s\n", format(round(subtotal),          big.mark = "
cat(sprintf("Contingency (%.0f%%)    : %s\n", contingency_r * 100,
            format(round(contingency), big.mark = " ")))
cat(sprintf("-----\nGRAND TOTAL      : %s\n",
            format(round(grand_total), big.mark = " ")))

```

How would top-down vs. bottom-up change this number?

The cell above is **bottom-up**: it sums the pieces and lets the total fall out. A **top-down** pass would start with a figure Senior Management is willing to fund — say “\$650k, no more” — and force each line to fit. In practice you do both: build bottom-up for defensibility, compare to the top-down ceiling, and negotiate the gap before the Charter is signed. If bottom-up comes in well over the ceiling, that is a **feasibility** finding, not a rounding problem.

9.11 Key Takeaways

- Cost management starts in **planning**. Sources: labour, materials, sub-contractors, equipment/facilities, travel, shipping.
- Every cost carries **two or three** of the eight labels: direct/indirect, fixed/variable, normal/expedited, recurring/non-recurring. **Direct labour cost = rate × hours**; indirect is often **20–50 % of direct**.
- Estimating method follows project stage: **ballpark ±30 %**, **parametric ±15 %**, **feasibility ±10 %**, **definitive ±5 %** — the **cone of uncertainty**. Always quote a **range**, and keep “estimate” separate from “commitment.”
- Estimate quality is driven by planning horizon, duration, people, structure, culture, and **padding** — which compounds into a meaningless total. Remove pads from work packages; hold one visible contingency.
- **Top-down** imposes discipline but breeds friction; **bottom-up** (via **activity-based costing** on the WBS) forces real planning. ABC = identify activities → cost drivers → rate per unit → rate × units.
- A **time-phased budget** produces the **cumulative cost baseline (S-curve)** you track against later.
- **Contingency reserve** (10–15 %, PM-owned, inside the baseline, covers known-unknowns) is distinct from the **management reserve** (sponsor-owned, outside the baseline, covers unknown-unknowns). Contingency is added **after** ABC, never inside it.

9.12 Practice Problems

Problem 1: Fully-burdened labour and an indirect-rate swing

A test campaign needs 2 technicians for 3 weeks (37.5 h/week each) at \$34/h,

plus \$6 200 of consumables. Compute the fully-burdened cost at an indirect rate of 25 %, and again at 45 %. How much does the indirect-rate assumption alone move the number?

Solution.

```
techs <- 2; weeks <- 3; hrs_week <- 37.5; rate <- 34
consumables <- 6200

direct_labour <- techs * weeks * hrs_week * rate
direct_cost <- direct_labour + consumables

for (ir in c(0.25, 0.45)) {
  total <- direct_cost * (1 + ir)
  cat(sprintf("Indirect %2.0f%%: direct $%s + indirect $%s = $%s\n",
             ir * 100,
             format(direct_cost, big.mark = " "),
             format(round(direct_cost * ir), big.mark = " "),
             format(round(total), big.mark = " ")))
}
cat(sprintf("\nSwing from the indirect-rate assumption alone: $%s\n",
           format(round(direct_cost * (0.45 - 0.25)), big.mark = " ")))
```

```
Indirect 25%: direct $13 850 + indirect $3 462 = $17 312
Indirect 45%: direct $13 850 + indirect $6 232 = $20 082
```

Swing from the indirect-rate assumption alone: \$2 770

The direct cost is fixed at \$21 500, but the burdened total moves by \$4 300 — 20 % of direct — purely on the overhead assumption. Always state the indirect rate you used.

Problem 2: Which estimating method, and what range?

For each situation, name the method and give the $\pm$ band around a \$500 000 point estimate.

- A retailer asks, in a hallway, roughly what a new product line will cost. You have a concept sketch.
- You have run three near-identical RC-vehicle launches in the last four years with clean cost records.
- All major POs are drafted and 90 % of the design is released for a firm-fixed-price defense bid.

Solution.

- (a) **Ballpark**, ± 30 % $\rightarrow$ \$350 000 – \$650 000.
- (b) **Comparative/parametric**, ± 15 % $\rightarrow$ \$425 000 – \$575 000.
- (c) **Definitive**, ± 5 % $\rightarrow$ \$475 000 – \$525 000. (Anything looser is a bad idea on a fixed-price bid.)

Problem 3: Variance analysis on the ABC budget

Using the preliminary ABC budget above (total planned \$63 250), the project finishes with these actuals: Survey \$4 250, Design \$8 000, Clear site \$3 500, Foundation \$8 500, Framing \$11 250, Plumb & wire \$5 150. Find the variance per activity and overall, and flag the activities that need a post-mortem.

Solution.

```
planned <- c(Survey = 4000, Design = 8000, `Clear site` = 4000,
             Foundation = 7500, Framing = 10000, `Plumb & wire` = 5000)
actual  <- c(4250, 8000, 3500, 8500, 11250, 5150)

variance <- planned - actual          # + = under budget
tab <- data.frame(Activity = names(planned),
                  Planned = as.integer(planned),
                  Actual = as.integer(actual),
                  Variance = as.integer(variance),
                  Pct = round(100 * variance / planned, 1))
print(tab, row.names = FALSE)

cat(sprintf("\nTOTAL planned %$s  actual %$s  variance %$s (%.1f%%)\n",
            format(sum(planned), big.mark = " "),
            format(sum(actual), big.mark = " "),
            format(sum(variance), big.mark = " "),
            100 * sum(variance) / sum(planned)))
```

Activity	Planned	Actual	Variance	Pct
Survey	4000	4250	-250	-6.2
Design	8000	8000	0	0.0
Clear site	4000	3500	500	12.5
Foundation	7500	8500	-1000	-13.3
Framing	10000	11250	-1250	-12.5
Plumb & wire	5000	5150	-150	-3.0

TOTAL planned \$38 500 actual \$40 650 variance \$-2 150 (-5.6%)

Design and (roughly) Survey held. **Foundation (-13 %)** and **Framing (-13 %)** blew through and drove the whole project \$2 150 (3.4 %) over — those two get the post-mortem.

Problem 4 (discussion): sizing contingency for the RC-vehicle project

The project brief flags several live risks: one candidate motor supplier is unproven, that supplier's staff are negotiating a first labour contract, the carbon-fibre supplier is mid-restructuring after an acquisition, and a new assembly line would carry a learning curve. The subtotal budget is \$620 000.

Would you size the contingency at 10 %, 15 %, or 20 %? Would you spread it

across all tasks or ring-fence it for specific ones? Defend your answer.

Discussion points. With that many concentrated, plausible risks, **10 % is light**; 15 % is defensible and matches the course's Costs Chart; 20 % is arguable if the unproven-supplier and new-line risks are both realised. A strong answer **ring-fences** most of the reserve against the supplier and assembly-line line items (the tasks critical to success) rather than smearing it evenly, and states that split in the scope statement so stakeholders are not surprised when it is drawn down.

9.13 Further Resources

- **Cost estimating techniques (analogous, parametric, bottom-up):** curated explainer videos below.
- Preview of Chapter 12 — Project Control Systems: the S-curve you built here becomes the baseline for **earned value** analysis.

The Basics of Project Cost Management (ProjectManager): <https://www.youtube.com/watch?v=EyPFi0YO32M>

Cost Estimates - Key Concepts in Project Management (David McLachlan): <https://www.youtube.com/watch?v=hfn9VCAj4Dw>

How to Estimate Project Costs Accurately: <https://www.youtube.com/watch?v=2IxoWVjsphc>

What Is Earned Value? (Ricardo Vargas) - looking ahead to project control: <https://www.youtube.com/watch?v=x0EgZnXdunw>

Chapter 10

Negotiation and Conflict Management

Note

This chapter is in preparation. It will be built from **Lecture 10 – Negotiation and Conflict Management** of MGMT-3105 (and its answer deck), following the same structure as Chapter 1 and Chapter 9.

10.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Explain why conflict is normal on projects and distinguish functional from dysfunctional conflict.
- Apply the Thomas-Kilmann modes (competing, collaborating, compromising, avoiding, accommodating) to a situation.
- Prepare for a negotiation: interests vs. positions, BATNA, ZOPA, reservation and target points.
- Apply principled negotiation (separate people from problem; focus on interests; invent options; use objective criteria).
- Manage conflict with suppliers, functional managers, and team members.
- Recognise cultural and ethical dimensions of negotiation in a global supply chain.

10.2 Topic outline

1. Sources of project conflict across the life cycle.

2. Conflict-handling modes and when each is appropriate.
3. Distributive vs. integrative negotiation.
4. Preparation: BATNA, ZOPA, anchoring.
5. Principled negotiation.
6. Difficult conversations and escalation paths.
7. Running case: negotiating the motor/control-system supply contract for the RC-vehicle project.

Chapter 11

Resource Management

Note

This chapter is in preparation. It will be built from **Lecture 11 – Resource Management** of MGMT-3105, following the same structure as Chapter 1 and Chapter 9.

11.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Distinguish time-constrained from resource-constrained scheduling.
- Build a resource histogram and identify over-allocation.
- Apply resource levelling and resource smoothing, and explain their effect on the critical path and float.
- Explain the theory-of-constraints view of a project (critical chain, buffers).
- Plan for procurement of materials and outsourced work.
- Manage the sustainability dimension of resources: material efficiency, energy, and waste.

11.2 Topic outline

1. Types of project resources: labour, equipment, material, facilities.
2. Resource loading and the resource histogram.
3. Over-allocation and the levelling problem.
4. Resource levelling vs. smoothing; splitting and multitasking costs.
5. Critical chain and buffer management.
6. Procurement and make-or-buy.

7. Sustainable resource use: circular-economy principles applied to the project.
8. Running case: levelling the RC-vehicle project team and tooling.

Chapter 12

Project Control Systems

i Note

This chapter is in preparation. The MGMT-3105 course map lists a *Control Systems* lecture in this slot; the chapter will be built from project-control fundamentals and the course's Project Manager Report and Team Member Report templates, following the same structure as Chapter 1 and Chapter 9.

12.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Describe the monitoring-and-control loop: measure, compare to baseline, analyse variance, act.
- Compute earned-value metrics: PV, EV, AC, SV, CV, SPI, CPI.
- Forecast with EAC, ETC, and TCPI, and interpret the results.
- Build and read an S-curve of planned vs. earned vs. actual cost.
- Run integrated change control and keep the baseline meaningful.
- Report status honestly: the “watermelon” trap and leading vs. lagging indicators.

12.2 Topic outline

1. What “control” means: closed-loop management.
2. The performance measurement baseline (scope + schedule + cost).
3. Earned value management: the three curves and the six core metrics.
4. Variance analysis and forecasting (EAC scenarios, TCPI).

5. Change control and configuration management.
6. Status reporting, dashboards, and honest metrics.
7. Running case: an earned-value snapshot of the RC-vehicle project at week 8.

Chapter 13

Project Closeout and Termination

Note

This chapter is in preparation. It will be built from **Lecture 12 – Project Closeout and Termination** of MGMT-3105, following the same structure as Chapter 1 and Chapter 9.

13.1 Learning objectives (planned)

By the end of this chapter you will be able to:

- Distinguish the ways a project ends: completion, extinction, addition, integration, starvation.
- Run an orderly administrative closeout: deliverable acceptance, contract closure, financial closure.
- Release resources and transition the product to operations or the customer.
- Facilitate a lessons-learned review and write to the organizational knowledge base.
- Explain the organizational and human implications of closing a project (team dispersal, morale).
- Recognise when to terminate a troubled project early, and how to do it responsibly.

13.2 Topic outline

1. Types of project termination.

2. The closeout checklist: scope verification, punch list, sign-off.
3. Contract and procurement closure.
4. Financial closure and final cost accounting.
5. Lessons learned and the post-implementation review.
6. Resource release and team transition.
7. Early termination: decision criteria and stakeholder communication.
8. Running case: closing out the RC-vehicle prototype project and handing off to manufacturing.

Bibliography